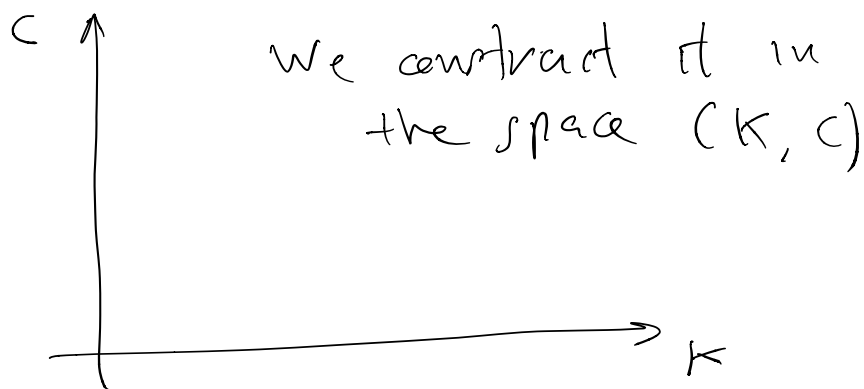


Notes on Continuous Time Model

Dynamic Macro II
Prof. Felipe Meza
August 2020

We will construct a phase diagram
I.e we will construct a set of points representing the behavior of a differential equation
(a locus)



Let us take first the Euler equation:

$$\dot{c}/c = \frac{1}{\sigma} (f'(K) - \delta - \rho)$$

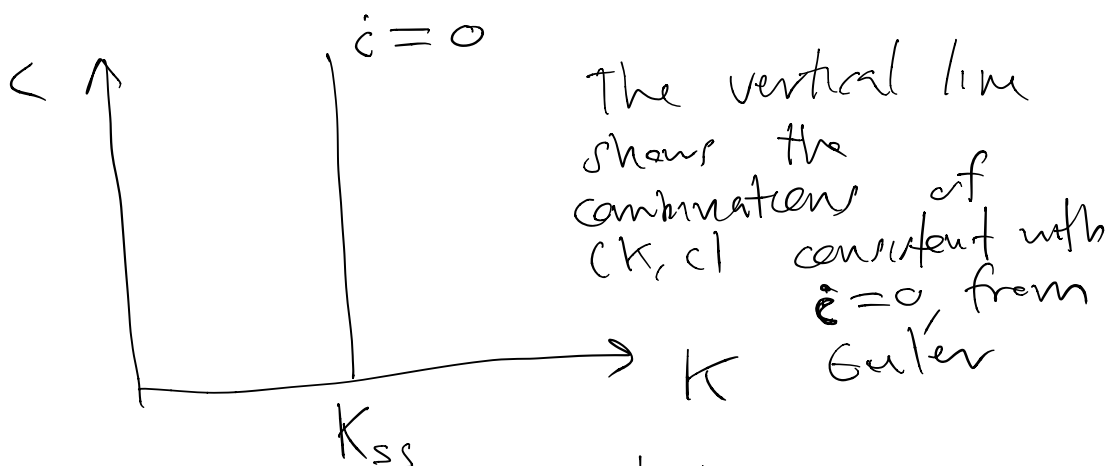
Set it to zero

$$\Rightarrow f'(k) = \delta + \rho$$

This equation depends only on k , not on c

Plus, we can see it produces a solution for k . That solution is its steady state value, i.e. when the economy stops moving

How does it look in the graph?



This is the steady state value for capital

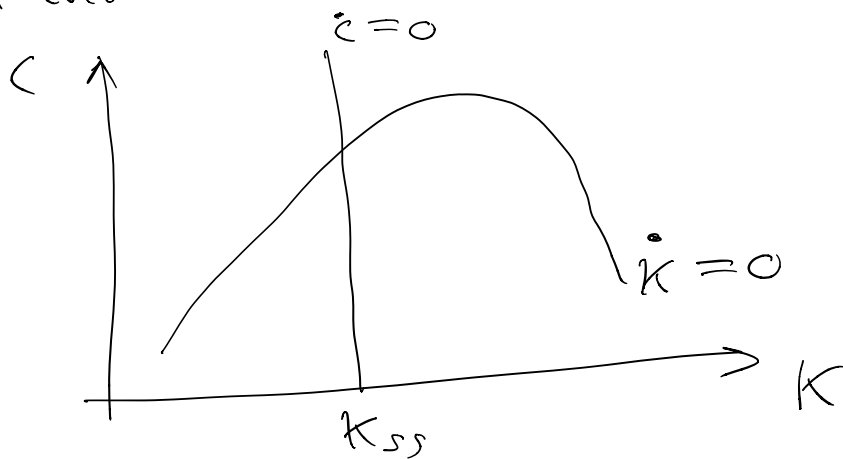
Let us now look at the other equation, feasibility

$$\dot{k} = f(k) - c - \delta k$$

Set it to zero

$$\dot{c} = f(k) - \delta k$$

The shape of the function depends on $f(k)$. Plus, with "high" levels of capital and decreasing marginal returns, eventually $-\delta k$ drives the function down

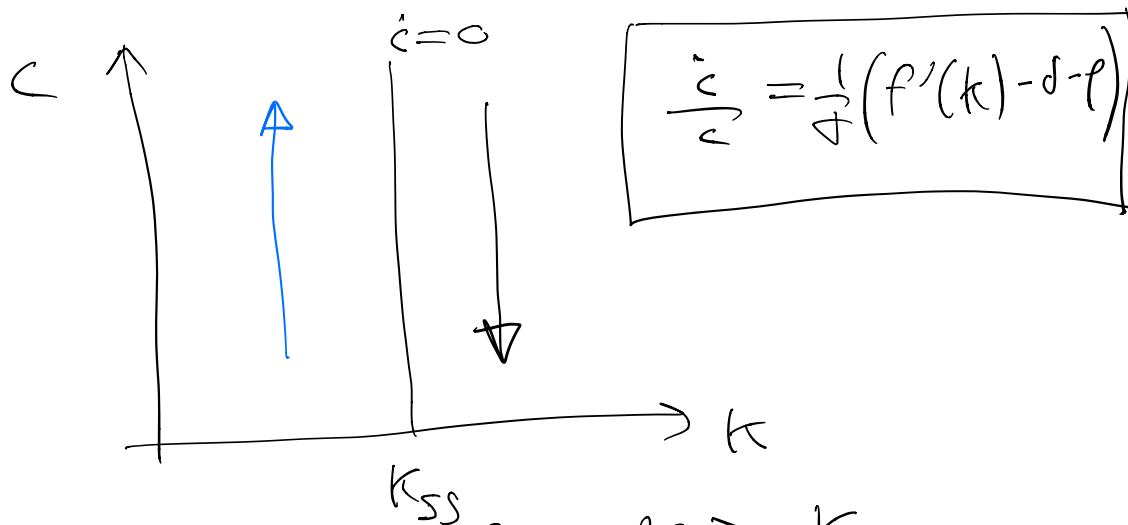


We can show (below...) that the maximum of $(\dot{k} = 0)$ is located to the right of k_{ss}

Dynamics: away from each locus, we have $\dot{c} > 0$ or $\dot{c} < 0$, and same for $\dot{k}$

The dynamics come from the behavior in each equation

Let's look first at $\dot{c} = 0$



With values for $k > k_{ss}$,
the marginal product $f'(k) < \delta + \rho$
Why? Because we have marginal
decreasing returns

$\Rightarrow \dot{c} < 0$
I.e. consumption falls over time

The behavior is the opposite
with $k < k_{ss}$

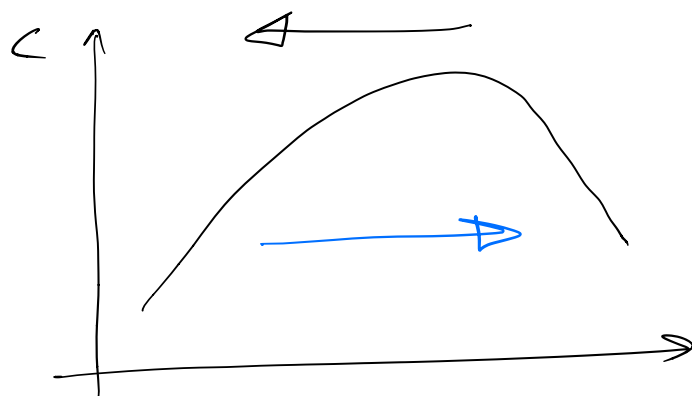
I.e. $\dot{c} > 0$

Now let's look at
 $\dot{k} = f(k) - c - \delta k$, away from locus $\dot{k} = 0$

Fix a value for k
(you can do it the other way around)

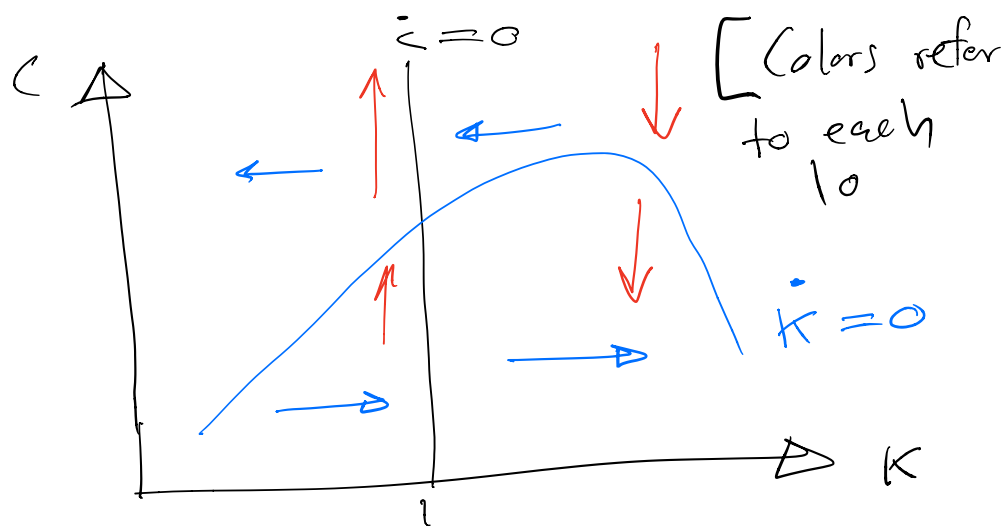
Then, what happens with $\dot{k}$ if c is higher?

k falls, i.e. $\dot{k} < 0$

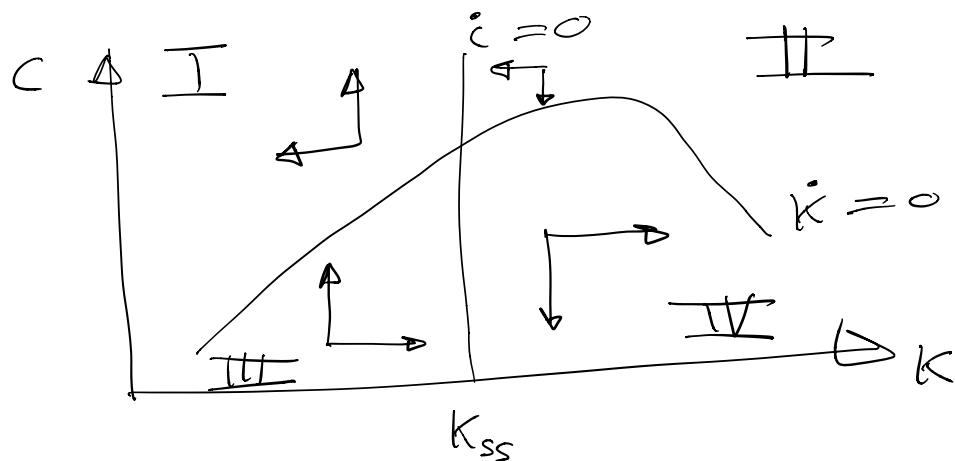


If c is smaller, then k increases over time, i.e. $\dot{k} > 0$

Let's combine all this information into a single graph



We summarize it like this



The arrows show the dynamics of (k, c) over time

In regions II and III, the dynamics lead us to the point in which $(\dot{k} = 0, \dot{c} = 0)$, i.e. to the steady state

Starting from a given $k(0)$, there is a trajectory that leads us to the steady state

→ Notice that we need a specific $c(0)$!

That is why this graph shows how the Shooting Method works

